

From Indexing Reviews to Indexing Evaluations: Ontology-Guided Construction of a Chinese Academic Review Index

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Abstract

Expert judgment about scholarship exists in written form and at scale – in review articles, scholarly commentaries, and book reviews – but it cannot be retrieved as judgment. Attempts to index the review literature date to the 1970s and were abandoned; they indexed reviews as documents and never recorded what those reviews evaluated. This dissertation treats the evaluative literature as a problem of knowledge organization. It develops a three-layer academic evaluation ontology of objects, dimensions, and viewpoints, anchored on the object layer; formalizes tacit disciplinary distinctions as executable boundary rules; and extracts evaluative statements through a staged, ontology-constrained workflow. On a 200-document reference corpus the full system reaches 87.34% entity F1 and 79.78% strict typed F1. Ablation separates the mechanisms: boundary rules govern type correctness, staging governs recall, and domain-adapted models prompted directly fall below 19% recall. The resulting Chinese Academic Review Index covers 13,906 reviews from 22 journals across 1900–2024, holding 115,451 evaluative sentences and 263,898 disambiguated objects typed by a nine-class evidence facet. At index scale, evaluation objects prove to carry multiple ontological identities: one term is attested under six distinct types.

Keywords

scholarly review literature; evaluation ontology; research assessment; knowledge organization; index construction

ACM Reference Format:

Yu Zhu. 2026. From Indexing Reviews to Indexing Evaluations: Ontology-Guided Construction of a Chinese Academic Review Index. In *Proceedings of ACM/IEEE Joint Conference on Digital Libraries 2026 – Doctoral Consortium (JCDL '26)*. ACM/IEEE Joint Conference on Digital Libraries, Frisco, TX, USA, 5 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

For a decade, research assessment reform has urged that expert judgment rather than metrics carry the weight of quality claims [10, 20]. Such judgment already exists in written form and at scale

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JCDL '26, TBD

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ACM ISBN 978-1-4503-XXXX-X/2026/06
<https://doi.org/XXXXXXX.XXXXXXX>

– in review articles, scholarly commentaries, and book reviews – but it cannot be retrieved *as judgment*. One can find the reviews; one cannot ask what the field has concluded about a given theory, method, or system.

This is not a new observation. In 1974 the Institute for Scientific Information launched the *Index to Scientific Reviews* (ISR), a multidisciplinary index to the review literature. Its founder expected it to answer an obvious need; nine years later he wrote that he was still wondering why it had not found an audience, and ISR was redesigned, its citation index replaced by a research-front specialty index [8]. Its companion, the *Index to Book Reviews in the Sciences*, launched in 1979 and was discontinued in 1981 for want of subscribers, even as readers kept asking ISI to revive it [7]. The need was articulated by the founder of citation indexing, attempted twice, and abandoned twice.

The reason is instructive. ISR offered an author index, a permuted title-word index, a corporate index, and a citation index – four ways of locating a review as a document, none of which records what the review evaluates or what judgment it reaches. In the humanities, Gale's *Contemporary Literary Criticism* [4] solved the other half: published since 1973, it organizes excerpted criticism by the work criticized, making the evaluated object the entry point, but it is editorially compiled, confined to literary criticism, and not queryable at the level of the individual statement. One tradition scaled and indexed the container; the other anchored on the object and could not scale. The constraint was labor, and the gap it left is still open: OpenAlex, Web of Science, and current research information systems index works, authors, institutions, and citation relations, but none of them represents what a piece of scholarship was judged to be.

This dissertation treats the evaluative literature as a problem of knowledge organization. It develops an academic evaluation ontology in three layers – evaluation objects, dimensions, and viewpoints – and uses the object layer as the anchor to which the other two attach. On this basis it builds the Chinese Academic Review Index (CARI), whose demonstration collection covers library and information science: 13,906 review articles from 22 journals spanning 1900–2024, yielding 115,451 evaluative sentences. The index and a retrieval prototype are deployed on an institutional server, currently restricted to the campus network, with public release planned for early 2027.

One finding shapes the argument that follows. At index scale, evaluation objects carry multiple ontological identities: the term *digital library* is indexed under six distinct types, predominantly as an information system (743 evidence snippets) and as a concept

(626), and marginally as a subfield, a phenomenon, an organization, and a developmental stage. A surface term denotes different evaluated objects in different evaluative contexts. This is why an index built on title words, of the kind ISR provided, could not serve evaluative retrieval – and why a typed object layer is not an embellishment on such an index but its condition of possibility.

2 Research Questions and Current Stage

RQ1. What type system represents evaluation objects in scholarly reviews with boundaries stable enough for consistent annotation and automated assignment?

RQ2. How can evaluative statements be extracted and indexed at scale while preserving traceability from each indexed claim to its source?

RQ3. Given a typed object layer and a typed evidence layer, what should the unit of retrieval be, and how should the utility of the resulting index be evaluated?

RQ1 and RQ2 are substantially answered. The ontology has been specified and validated against a manually reviewed reference corpus; the extraction framework has been implemented and evaluated against rule-based, zero-shot, and ablated baselines; and the demonstration collection has been built, disambiguated, and deployed behind a retrieval interface. What remains open is not whether the layers can be built but how they should be composed for retrieval, and against what standard an evaluation index should be judged useful. The author is in the second year of a doctoral program in information resources management and expects to complete the degree in 2029.

3 Background and Related Work

Scholarly reviews are not summaries with judgment attached; judgment is the constitutive act of the genre, and studies of review articles and academic book reviews have shown that they stabilize knowledge foundations, confer legitimacy on emerging fields, and negotiate criticism through conventionalized rhetorical moves [2, 21, 26]. Research evaluation has begun to treat such judgment as data: F1000 recommendations have been compared with citation indicators, and open peer review has opened reviewer reasoning to analysis [3, 22, 23]. These studies, however, draw on platforms that were already structured at source. The far larger body of judgment carried in conventional reviews, commentaries, and book reviews remains unstructured.

The infrastructures that might hold it do not model it. The Functional Requirements for Bibliographic Records (FRBR) and the Bibliographic Ontology (BIBO) describe bibliographic entities, VIVO and the Common European Research Information Format (CERIF) describe researchers and research activities, and OpenAlex indexes works, authors, institutions, and concepts at scale [5, 6, 12, 14, 18]. All of them answer what exists and how it is connected; none answers what was judged, on what grounds, or with what verdict.

Existing extraction research has largely focused on a different genre. Domain-adapted encoders and document-level extraction datasets target research articles, where entities are concrete and typically named [1, 13, 16]. Work on cross-domain recognition reports that models continue to confuse fine-grained entities with the higher-level concepts that subsume them [17] – precisely the failure

mode that dominates evaluative text, where what is evaluated is often a theory, a methodology, or a disciplinary turn rather than a named object. No existing type system defines these objects, and no extraction pipeline is constrained by one.

4 Research Approach

4.1 An ontology of evaluation

The academic evaluation ontology is organized in three layers. Evaluation objects answer what is evaluated; evaluation dimensions specify the grounds; evaluation viewpoints record the judgment with its polarity and supporting evidence. Objects are the anchor: dimensions and viewpoints attach to them, so their type boundaries determine whether anything above them can be indexed consistently.

Objects are organized under four top-level classes – Agent, Artifact, Abstract entity, and Event – following Guarino’s [9] distinction between endurants and perdurants, with artifact hierarchies informed by FRBR, organizational facets by Ranganathan’s [19] faceted classification, and abstract knowledge objects by Kuhn [15] and Hjørland [11]. Compatibility with OpenAlex entity types is preserved deliberately, so that an evaluation layer can eventually be attached to existing scholarly knowledge infrastructures rather than standing apart from them.

The decisive design element is not the taxonomy but the boundary rules. Evaluative texts are dense with distinctions that disciplinary readers apply tacitly and models do not: methodology against method against technique; theory against model against framework; policy documents against policy initiatives; databases evaluated for coverage against information systems evaluated for function; organizations classified by role in knowledge production rather than by name. Five such confusion groups are formalized as explicit inclusion and exclusion criteria and applied identically in annotation and in automated assignment.

A single case shows what formalization buys. *Content analysis* appears in review texts sometimes as an overarching research stance, sometimes as a reusable procedure, and sometimes as a specific coding operation, and the three readings belong to three different types. The rule does not resolve this lexically; it directs the decision to the object’s function in the evaluative claim – whether the review is appraising a stance toward evidence, a procedure others may adopt, or an operation performed in one study. The same discipline of decision is what a human annotator applies without articulating it, and what a model, absent the rule, does not apply at all. Where no stable assignment can be reached even so, the object is retained under a residual label with its supporting evidence and the documented source of uncertainty, so that boundary cases accumulate as material for revising the ontology rather than disappearing into a default.

4.2 Ontology-constrained extraction

Extraction is decomposed into three agents operating on a single backbone model. The first performs high-recall candidate discovery, tolerating over-generation because omission is the costlier error at that stage. The second validates candidates against the ontology and assigns hierarchical types under the boundary rules. The third reviews the typed output for boundary drift and insufficient

evidence. Staging a single generative act into inspectable intermediate states isolates omission, misclassification, and unjustified assignment, making them separately detectable and correctable.

5 Preliminary Results

5.1 Extraction performance

A reference corpus was built from 200 Chinese library and information science review articles indexed in the Chinese Social Sciences Citation Index (CSSCI), selected for high evaluative density in order to exercise the type boundaries under load. It contains 25,859 entities across 5,557 evaluative sentences. A 1.5% random sample (388 entities) was independently reviewed by two researchers; agreement between reviewers was $\kappa = 0.833$, and agreement between each reviewer and the corpus exceeded 0.90.

Performance was measured against two baselines and two ablations. The baselines are a rule-based extractor built from lexical patterns and type keywords (B1), and the same backbone large language model (LLM) prompted zero-shot with the type inventory but with neither boundary rules nor staging (B2). The ablations withdraw one mechanism each from the full system: A1 collapses the three agents into a single generative pass while retaining the boundary rules, and A2 retains the staged workflow but withdraws the boundary rules, leaving only type names and short glosses.

The two mechanisms are separable and act on different quantities. Removing the boundary rules while retaining the staged workflow (A2) costs 23.2 points of strict typed F1 and 17.8 points of type accuracy: without executable boundaries, the system finds the right spans and assigns the wrong kinds. Removing the staging while retaining the rules (A1) costs 19.7 points of recall: without inspectable intermediate states, candidates are lost inside a single generative act and cannot be recovered downstream.

Table 2 localizes the first effect. Confusion is measured within each of the five groups the boundary rules were written to separate, so the quantity reported is the rate at which a type is exchanged for its nearest neighbour rather than for an unrelated type.

The effect is concentrated where the distinction is most abstract. Methodology, method, and technique differ by the level at which a practice is being appraised rather than by any lexical marker, and it is there that removing the rules multiplies confusion sixfold; database against information system, a distinction with observable correlates in the text, barely moves. This is the argument for treating boundary formalization as ontology work rather than as prompt engineering: what the rules supply is not additional instruction but the disciplinary discrimination that the text leaves implicit.

Model scale alone does not appear sufficient for this task. Two domain-adapted Chinese models prompted directly under the same ontology reached recall of 18.61% (LISGPT-14B [25]) and 14.71% (HssaLLM-32B [24]). Domain knowledge without staged constraint does not locate evaluation objects.

5.2 The resulting index

The demonstration collection covers library and information science: 13,906 review articles from 22 journals published between 1900 and 2024, yielding 115,451 evaluative sentences and 263,898 distinct evaluation objects after deduplication and disambiguation. Journals were selected for sustained publication of review and

book-review content in the field, and the long span is a property of the source titles rather than a target: coverage is thin before the 1980s and dense thereafter, which makes the collection usable for diachronic questions only with that caveat stated.

An index record is anchored on the evaluation object rather than on the reviewing document. Each record carries the object's preferred label and variants merged during disambiguation, its full ontology path from top-level class through facet to type label, the evidence snippets that attest it, and, for every snippet, the evidence type and a resolvable link to the source record. Because a term may be attested under more than one type, records are keyed on the object-type pair rather than on the string, and the interface exposes the alternative identities of a term alongside whichever one is in view. The consequence that matters for research assessment is decomposability: any aggregate the index produces about an object can be taken apart into the individual statements that produced it, and each of those statements returned to the article and the sentence in which it was made.

Snippets are typed by a nine-class evidence facet – direct appraisal, limitation, normative advocacy, classificatory synthesis, trend judgment, causal explanation, comparative judgment, hedged judgment, and research gap – which implements the viewpoint layer for retrieval purposes. The classes are not a sentiment scale. They record what kind of evaluative act a statement performs, on the reasoning that a limitation and a research gap point in different directions for an evaluator even when both read as negative, and that hedged judgment is evidence about the state of a field's consensus rather than a weakened version of appraisal.

Table 3 records the finding announced in Section 1, now with its distribution. Six identities are attested for a single term, and two of them carry 95% of the evidence: the field evaluates the digital library overwhelmingly as a system to be assessed for function and as a concept to be assessed for adequacy, and only marginally as a subfield, a phenomenon, an organization, or a stage. The observed multiplicity is consistent with the contextual nature of evaluative discourse rather than with a single fixed identity for each surface term.

6 Challenges and Questions for the Consortium

Three lines of work remain. The dimension layer, which records the grounds on which an object is evaluated, is specified but not yet extracted at collection scale. The source base is confined to reviews and book reviews published in journals, and extending it to monographs, peer review reports, and post-publication commentary will test whether the type system holds across genres with different evaluative conventions. And the collection itself is confined to library and information science, so the transferability of the top-level classes to other humanities and social science fields is asserted by design but not yet demonstrated.

The dissertation is planned in six chapters. Two are complete in substance as reported above. A third, the construction and disambiguation of the demonstration collection, is complete as an artifact and awaits written analysis. The remaining three – the dimension layer at collection scale, the retrieval design and its evaluation, and the implications for evidence-based research assessment – are

Table 1: Extraction performance on the test set (%).

Setting	Precision	Recall	Entity F1	Strict typed F1	Type acc.
B1 rule-based	40.93	20.67	27.47	13.10	47.68
B2 zero-shot large language model	73.07	65.07	68.84	46.99	68.26
A1 – staging	82.89	64.81	72.74	57.41	78.92
A2 – boundary rules	80.08	74.03	76.94	56.60	73.57
Full	90.33	84.55	87.34	79.78	91.35

Table 2: Intra-group confusion rate by ablation setting (%).

Confusion group	Full	A1 – staging	A2 – boundary rules
Methodology / method / technique	4.11	9.95	25.86
Research / service / professional / governance organization	2.13	3.61	5.44
Theory / model / framework	0.50	1.10	4.68
Database / information system	0.87	1.81	1.21
Policy document / policy initiative	0.46	1.14	1.82
Overall	0.68	1.49	3.40

Table 3: Ontological identities of one surface term (*digital library*) in the demonstration collection.

Top-level class / facet	Type label	Records	Snippets
Artifact / Functional artifact	Information system	280	743
Abstract entity / Conceptual	Concept, term	282	626
Abstract entity / Epistemic community	Subfield	27	33
Abstract entity / Phenomenal	Phenomenon	29	30
Agent / Organization	Professional service organization	8	9
Event / Process	Developmental stage	3	3

where the questions below fall, and where the argument is still open to being made differently.

Three questions are genuinely open, and are the ones the author hopes to discuss.

What is the unit of retrieval? An object-anchored index can return objects, evidence snippets, or object–dimension–viewpoint triples. The triple is the most precise and the most decontextualized: a judgment separated from the argument that produced it may be accurate and still misleading. Consider a query that would be impossible against a citation index – the limitations the field has attributed to digital libraries *as systems*, excluding the parallel literature that evaluates the term as a concept. The index can compose that query today, from the type facet and the evidence facet together. What it cannot yet say is what should be returned: the objects that satisfy it, ranked somehow; the snippets themselves, which are quotable but stripped of their argumentative surroundings; or a synthesis, which is useful and unattributable. A scholar tracing how a concept was received, an evaluator assembling evidence about a body of work, and a bibliometrician studying evaluative discourse plausibly want different answers, and the index cannot be primary for all three.

How is an evaluation index validated without a gold standard? Extraction performance was measured against a reference corpus built under the same ontology that governs the system, which establishes internal consistency but not external validity. Candidate strategies

include task-based evaluation with domain experts, convergence testing against independent indicators of reception, and comparative retrieval studies against citation-based search. Each measures something different, and the choice among them effectively defines what the index is for.

How far do the top-level classes travel? The four classes rest on general ontological distinctions, but the type labels beneath them, and several boundary rules, were calibrated on the evaluative vocabulary of one discipline in one language. Which parts of the scheme are epistemically general and which are artifacts of that calibration is an empirical question the dissertation must answer before it can claim more than a demonstration.

Fifty years after the review literature was first indexed and the attempt abandoned, the constraint that defeated it has lifted. What was missing then was not computing power but a way to say what a review is about when what it is about is a judgment. That is the layer this dissertation builds.

Acknowledgments

The author thanks Prof. Jiyuan Ye who supervised this research.

This article is an outcome of the major project "Research on the Construction of a Characteristic Chinese Evaluation System for Philosophy and Social Sciences from the Perspective of All-round

Research Evaluation" (No. 24&ZD323) supported by the National Social Science Foundation of China, and the Postgraduate Research Practice Innovation Program of Jiangsu Province titled "Optimization of the All-round Research Evaluation of Philosophy and Social Sciences Assisted by Generative AI" (No. 26CXJH0170).

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